

```
In [3]: import pandas as pd
df = pd.read_csv('customer_shopping_behavior.csv')

In [4]: df.head()

Out [4]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

```
In [5]: df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Customer ID           3900 non-null   int64
1   Age                   3900 non-null   int64
2   Gender                3900 non-null   object
3   Item Purchased        3900 non-null   object
4   Category              3900 non-null   object
5   Purchase Amount (USD) 3900 non-null   int64
6   Location               3900 non-null   object
7   Size                  3900 non-null   object
8   Color                 3900 non-null   object
9   Season                3900 non-null   object
10  Review Rating         3863 non-null   float64
11  Subscription Status    3900 non-null   object
12  Shipping Type         3900 non-null   object
13  Discount Applied      3900 non-null   object
14  Promo Code Used       3900 non-null   object
15  Previous Purchases    3900 non-null   int64
16  Payment Method        3900 non-null   object
17  Frequency of Purchases 3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB

In [7]: df.describe(include='all')

Out [7]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

```
In [9]: df.isnull().sum()

Out [9]: Customer ID      0
Age      0
Gender    0
Item Purchased    0
Category      0
Purchase Amount (USD)  0
Location      0
Size      0
Color      0
Season      0
Review Rating    37
Subscription Status    0
Shipping Type    0
Discount Applied    0
Promo Code Used    0
Previous Purchases    0
Payment Method      0
Frequency of Purchases    0
dtype: int64

In [11]: df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))

In [34]: df.isnull().sum()
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   customer_id           3900 non-null   int64
1   age                   3900 non-null   int64
2   gender                3900 non-null   object
3   item_purchased        3900 non-null   object
4   category              3900 non-null   object
5   purchase_amount       3900 non-null   int64
6   location               3900 non-null   object
7   size                  3900 non-null   object
8   color                 3900 non-null   object
9   season                3900 non-null   object
10  review_rating         3900 non-null   float64
11  subscription_status    3900 non-null   object
12  shipping_type         3900 non-null   object
13  discount_applied      3900 non-null   object
14  promo_code_used       3900 non-null   object
15  previous_purchases    3900 non-null   int64
16  payment_method        3900 non-null   object
17  frequency_of_purchases 3900 non-null   object
18  age_group             3900 non-null   category
19  purchase_frequency_days 3353 non-null   float64
dtypes: category(1), float64(2), int64(4), object(13)
memory usage: 583.0+ KB

In [17]: df.columns = df.columns.str.lower()

df.columns = df.columns.str.replace(' ', '_')

df=df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
df.columns

Out [17]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'promo_code_used', 'previous_purchases',
      'payment_method', 'frequency_of_purchases'],
      dtype='object')

In [19]: #create a columns age_group
labels = ['Young Adult','Adult','middle-aged','senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)

In [20]: df[['age','age_group']].head(10)

Out [20]:
```

	age	age_group
0	55	middle-aged
1	19	Young Adult
2	50	middle-aged
3	21	Young Adult
4	45	middle-aged
5	46	middle-aged
6	63	senior
7	27	Young Adult
8	26	Young Adult
9	57	middle-aged

```
In [ ]:

In [32]: # create columns purchase_frequency_days
frequency_mapping= {
'Fortnightly':14,
'Weekly':7,
'Monthly':30,
'Quarterly':90,
'Bi_Weekly':14,
'Annually':365,
'Every 3 Months':90
}
df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)

In [33]: df[['purchase_frequency_days','frequency_of_purchases']].head(10)

Out [33]:
```

	purchase_frequency_days	frequency_of_purchases
0	14.0	Fortnightly
1	14.0	Fortnightly
2	7.0	Weekly
3	7.0	Weekly
4	365.0	Annually
5	7.0	Weekly
6	90.0	Quarterly
7	7.0	Weekly
8	365.0	Annually
9	90.0	Quarterly

```
In [35]: df[['discount_applied','promo_code_used']].head(10)

Out [35]:
```

	discount_applied	promo_code_used
0	Yes	Yes
1	Yes	Yes
2	Yes	Yes
3	Yes	Yes
4	Yes	Yes
5	Yes	Yes
6	Yes	Yes
7	Yes	Yes
8	Yes	Yes
9	Yes	Yes

```
In [36]: (df['discount_applied'] == df['promo_code_used']).all()

Out [36]: np.True_

In [39]: df = df.drop('promo_code_used', axis=1, errors='ignore')
df.columns

Out [39]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
      'purchase_amount', 'location', 'size', 'color', 'season',
      'review_rating', 'subscription_status', 'shipping_type',
      'discount_applied', 'previous_purchases', 'payment_method',
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
      dtype='object')

In [40]: pip install psycpg2-binary sqlalchemy

Requirement already satisfied: psycpg2-binary in c:\users\sistech computer\appdata\local\programs\python\python314\lib\site-packages (2.9.12)
Requirement already satisfied: sqlalchemy in c:\users\sistech computer\appdata\local\programs\python\python314\lib\site-packages (2.0.49)
Requirement already satisfied: greenlet>=1 in c:\users\sistech computer\appdata\local\programs\python\python314\lib\site-packages (from sqlalchemy) (3.5.1)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\sistech computer\appdata\local\programs\python\python314\lib\site-packages (from sqlalchemy) (4.15.0)
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 25.3 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip

In [44]: from sqlalchemy import create_engine

username = "postgres"
password = "yasir123"
host = "localhost"
port = "5432"
database = "customer_behavior"

engine = create_engine(f"postgresql+psycpg2://{username}:{password}@{host}:{port}/{database}")

table_name = "customer"

df.to_sql(table_name, engine, if_exists="replace", index=False)

print("Data uploaded successfully")

Data uploaded successfully

In [ ]:
```


